MILITARY SPECIFICATION

COMPUTER, FLIGHT DIRECTOR, AIRCRAFT

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - This specification covers certain design and all performance requirements for the procurement of one type of aircraft flight director computer.

1.2 Associated Equipment - The computer manufactured in accordance with this specification is a component of an Integrated Flight Instrumentation System conforming to Specification MIL-S-23535. The other components of the system are the Horizontal Situation Indicator manufactured in accordance with Specification MIL-I-23366 and the Remote Attitude Director Indicator manufactured in accordance with Specification MIL-I-23524.

2. APPLICABLE DOCUMENTS

2.1 General - The following documents, of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

BB-N-411	Nitrogen
QQ-C-320	Chromium Plating (Electrodeposited)
QQ-N-290	Nickel Plating (Electrodeposited)
QQ-P-416	Plating, Cadmium (Electrodeposited)

FSC 6605

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Military

MIL-P-116	Preservation, Methods of
MIL-M-3171	Magnesium Alloy, Processes for Corrosion Protection of
MIL-E-4682	Electron Tubes and Transistors, Choice and Application of
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-C-5541	Chemical Films for Aluminum and Aluminum Alloys
MIL-L-6085	Lubricating Oil, Aircraft Instrument, Low Volatility
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-S-6872	Soldering Process, General Specification for
MIL-R-7060	Radio Set AN/ARN-21
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series, General Specification for
MIL-M-7793	Meter, Time Totalizing
MIL-P-7936	Parts and Equipment, Aeronautical, Preparation for Delivery
MIL-R-8019	Receiver, Radio, R-541/ARN-14D
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-R-22973	Reliability Index Determination for Avionics Equipment Models, General Specification for
MIL-I-23366	Indicator, Horizontal-Situation
MIL-I-23524	Indicator, Attitude-Director, Remote
MIL-S-23535	Integrated Flight Instrumentation System

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MIL-R-25495	Radio Receiving Set AN/ARN-14
MIL-R-25497	Radio Receiving Set AN/ARN-18
MIL-C-26482	Connectors, Electric, Circular, Miniature, Quick Disconnect
MIL-D-70327	Drawings, Engineering and Associated Lists

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MS3116	Connectors, Plug, Electric, Straight, Miniature
MS33586	Metals, Definition of Dissimilar
MS17870	Computer, Flight Director, Aircraft

2.2 Availability of Documents - When requesting any of the applicable documents, refer to both title and number. Copies of this specification and applicable documents may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code DCS, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

3. REQUIREMENTS

3.1 Preproduction - The flight director computer furnished under this specification shall be a product, samples of which have been tested and have passed the preproduction tests specified herein and in the specification for the Integrated Flight Instrumentation System, Specification MIL-S-23535.

3.2 Selection of Government Documents - Specifications and Standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143 except as provided in 3.2.1 and 3.2.2.

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3.2.1 Commercial Parts - Commercial parts having suitable properties shall be used, where on the date of invitations for bids, there are not suitable standard parts. In any case, commercial utility parts, such as screws, bolts, nuts, cotter pins and having suitable properties, shall be used provided:

(1) They can be replaced by the standard parts (MS or AN) without alteration.

(2) The corresponding standard part numbers are referenced in the parts list and if practicable, on the contractor's drawings.

3.2.2 Standard Parts - With the exception of 3.2.1, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part number.

3.3 Materials - Materials shall conform to applicable specifications and shall be as specified herein. Materials for which there are no applicable specifications, or which are not specifically described herein, shall be of the best quality, of the lightest practicable weight and suitable for the purpose intended.

3.3.1 Critical Materials - Noncritical materials shall be used where practicable. Where the use of a critical material is essential to meet specification requirements, the material used shall be the least critical of those which are adequate for the purpose.

3.3.2 Non-magnetic Materials - Non-magnetic materials shall be used for all parts of the instrument except where magnetic materials are essential.

3.3.3 Protective Treatment - When materials are used in the construction of an instrument which is subject to corrosion in environmental conditions likely to occur during normal service life, they shall be protected against such corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification. Finishes and protective coatings which will crack, chip or scale during normal service life or are affected by extremes of environmental conditions, shall not be used.

3.3.3.1 Metals - Metals shall be of the corrosion resistant type, or shall be suitably protected as specified herein to resist corrosion due to fuels, salt spray or atmospheric conditions to which the instrument may be subject when in storage or during normal service life.

3.3.3.2 Dissimilar Metals - Dissimilar metals as defined in MS33586 shall not be used in intimate contact with each other, unless suitably protected against electrolytic corrosion by means of protective coatings or hermetic sealing.

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3.3.3.3 Magnesium Alloy Parts - Magnesium alloy parts shall be treated in accordance with Specification MIL-M-3171. When abrasion resistance is a factor, an anodic treatment approved by the procuring activity shall be used.

3.3.3.4 Alluminum Alloy Parts - Where practicable, alluminum alloy parts shall be covered with an anodic film conforming to Specification MIL-A-8625. Small holes, pipe threads and case inserts need not be anodized. Alluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with Specification MIL-C-5541.

3.3.3.5 Iron and Steel Parts - Iron and steel parts which are in hermetically sealed cases shall be cadmium plated in accordance with Specification QQ-P-416. Parts which are not in sealed cases shall be chromium or nickel plated in accordance with Specifications QQ-C-320 or QQ-N-290 respectively. Parts in a confined spare in the presence of organic material shall be tin-cadmium plated.

3.3.3.6 Fungous-Proof Materials - Materials which are nutrients for fungi shall not be used where it is practicable to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. If used in a hermetically sealed case, fungicidal treatment will not be necessary.

3.3.3.7 Fumes and Vapors - Materials used in the construction of the instrument shall not produce corrosive, deleterious or toxic fumes or vapors under the conditions specified herein.

3.3.4 Lubrication - The lubricating oil used in the instrument shall be in accordance with Specification MIL-L-6085. The amount of lubricating oil used shall be sufficient for at least 1000 hours of operation.

3.3.5 Workmanship - The instrument, including all parts and accessories, shall be so constructed and finished that it will be free from all defects which would affect proper functioning in service. Particular attention shall be given to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, welding and brazing, plating, painting, riveting, machine screw assemblies and freedom of parts from burrs and sharp edges.

3.3.5.1 Dimensions - Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation or performance of the instrument, they shall be held or limited accordingly.

3.3.5.2 Soldering - Soldering shall be performed in accordance with Specification MIL-S-6872.

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3.3.5.3 Screw Threads - Screw threads 0.060 inch in diameter or larger, shall be in accordance with Specification MIL-S-7742.

3.3.5.4 Electronic Parts - Electronic parts and the application thereof shall be in accordance with Specification MIL-E-5400. Parts that do not appear on approved lists in accordance with Specification MIL-E-5400 shall not be used unless approved by the procuring activity prior to use.

3.3.5.5 Electron Devices - Electron tubes, transistors and diodes shall be chosen and applied, and the complements reported as outlined in Specification MIL-E-4682. The complement report must be submitted to the procuring activity for review and approval prior to preproduction testing.

3.3.5.6 Cleaning - The instrument shall be thoroughly cleaned of loose, spattered or excess solder, resin flash that may crumble, metal chips and other foreign matter, after final assembly.

3.4 Design and Construction - The computer shall be designed to supply the proper horizontal steering pointer and vertical steering pointer and warning signals to a remote attitude director indicator to indicate to the pilot the proper control action during heading, command heading (data link), ILS intercept, ILS approach and radio track (VOR and/or TACAN) modes of manual automatic flight control system flight. During automatic navigation flight the computer shall bias all computed information out of view. The data link mode shall include submodes for traffic control, automatic carrier landing and precision course direction. During manual heading and radio track modes the vertical pointer shall display left-right steering information such that it is centered when correct control action is taken. When the low altitude bombing system (AJB-3A) is in operation the LABS computed steering signals in pitch and roll will go directly through the flight director computer to the vertical and horizontal pointers of the remote attitude director indicator. During ILS intercept the glide slope displacement signal and warning flag signals will go directly through the computer to the director indicator and the vertical pointer of the indicator shall display a preset intercept angle to the ILS localizer beam with asymptotic intercept of the localizer beam when correct centering control action is taken. During ILS approach mode the horizontal pointer of the indicator shall appear into view and the correct zeroing control action shall cause asymptotic intercept of the glide slope beam. During data link mode suitable steering information will be computed and displayed on the director indicator in accordance with the particular data link submode that is existent. The instrument shall be so constructed that no parts will work loose in service and will withstand the normal shocks, vibrations and such other conditions as are incident to shipping, storage, installation and service without failure.

3.4.1 Maintenance - The design shall be reasonably simple to facilitate as much as possible disassembly, repair or overhaul, service maintenance and reassembly by those tools and items of maintenance equipment which are normally available as commercial standards without requiring special tools and fixtures.

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3.4.2 Case - The outline dimensions of the case shall conform to MS17870. The case shall be made of non-ferrous, low density metal, uniform in texture, with a smooth surface and shall be / hermetically sealed. The case shall be finished with a lusterless black material, Color No. 37038 of FED-STD-No. 595. The finishing material shall be of a durable type to withstand usage encountered in service.

3.4.2.1 Hermetic Sealing - The case shall provide a hermetically sealed enclosure for all the mechanism except for the external adjustment which shall be gasket sealed under a dust cover. The case shall be so constructed that it may be opened, the mechanism removed and replaced and the case resealed at least three times. This shall be possible without the use of any special tool, jig or fixture, unless such device is specifically approved by the procuring activity. The sealing of the case shall not be dependent upon any material which will be adversely affected by any atmosphere to which the instrument may be subjected in normal use in military aircraft.

3.4.2.2 Filling Medium - The filling medium shall be a mixture of 90 percent nitrogen and 10 percent helium. The nitrogen used shall be of a quality in accordance with Specification BB-N-411, Type I, Class 1, Grade A. The filling medium shall be of at least 98 percent purity, free of dust particles and containing not more than 0.006 milligram of water vapor per liter (dew point -65°C) at the filling pressure. The case shall be filled to an absolute pressure of 1 $\pm$0.1 atmosphere.

3.4.2.3 Leak Rate - The computer shall be sealed such that the initial maximum leak rate at a pressure differential of approximately one atmosphere will not permit more than 10 percent loss of the total filling medium after 1000 hours.

3.4.2.4 Shockmount Base - The computer shall incorporate a shockmount base. This shockmount shall be designed so that the instrument operates properly when the complete computer assembly is subjected to the vibrations and accelerations anticipated during service usage. The shockmount base shall conform to MS17870. The base design shall be such that it can be removed from the computer case and reversed in direction with respect to the case.

3.4.2.5 Time Totalizing Meter - The instrument shall include within its case a time totalizing meter conforming to Specification MIL-M-7793.

3.4.2.6 Weight - The flight director computer shall be so designed that when completely assembled the weight shall not exceed 8.0 pounds.

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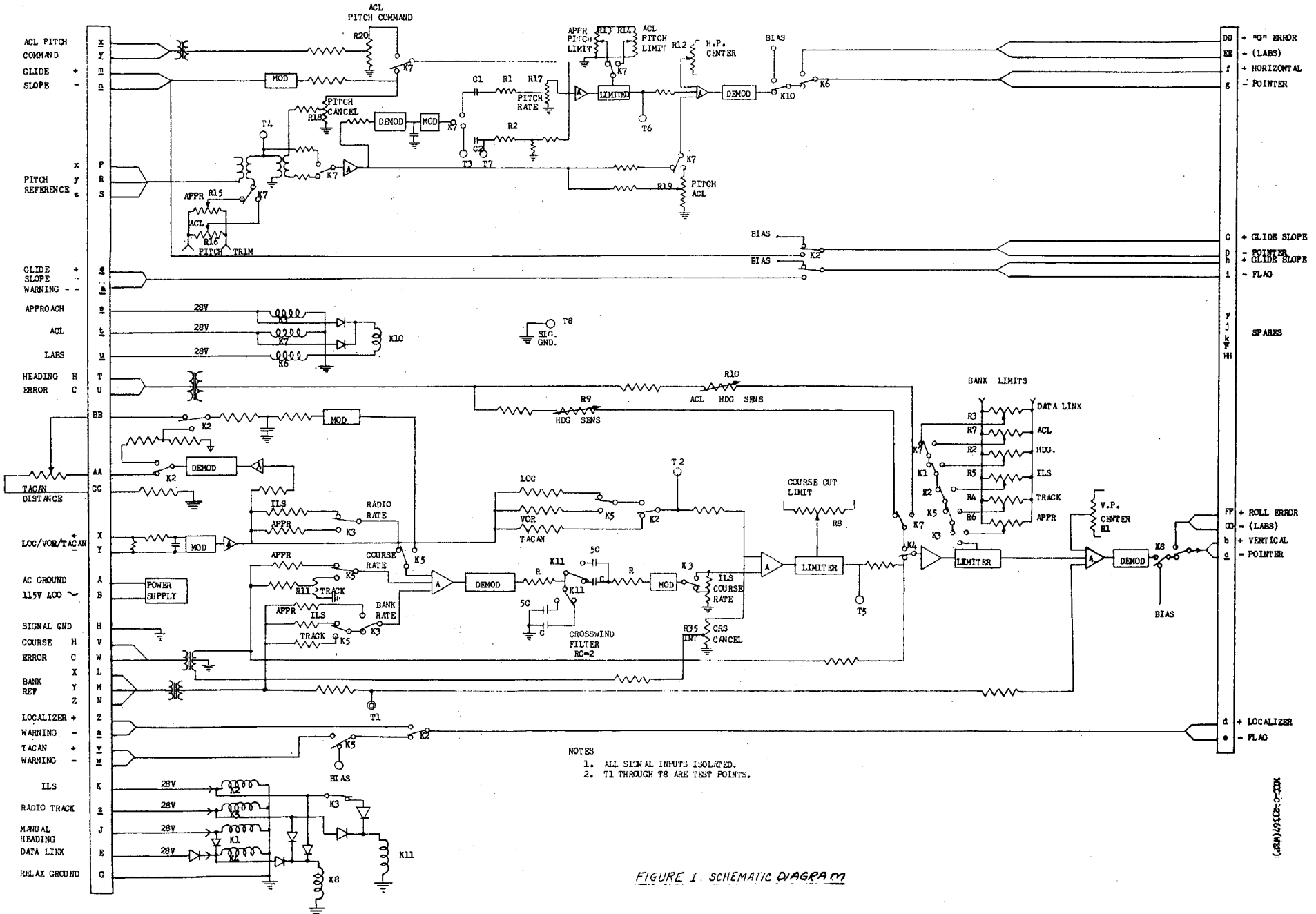


FIGURE 1. SCHEMATIC DIAGRAM

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3.4.3 Electrical Function Design - The computer shall operate as shown in the functional schematic diagram Figure 1.

3.4.3.1 Roll and Pitch Angle Signals - The roll and pitch angle input signals will be in the form of separate transmitter synchro threewire stator outputs in accordance with the standard test transmitter as specified herein with the dial readings set at 180 degrees for zero roll and pitch displacement. Increasing dial readings shall correspond to right roll and pitch-up maneuvers. Decreasing dial readings shall correspond to opposite maneuvers. The load placed on these signals within the computer shall be equivalent to high impedance, balanced loads which are equal to Eclipse-Pioneer Type AY500-5 control transformers which have their rotors connected to 10,000 ohm minimum loads. The load measured between any two lines of these inputs shall be at least 5000 ohms. The signal gradient input to the computer for roll and pitch angle signals shall be 200 millivolts per degree.

3.4.3.2 Heading and Course Error Signals - The heading and course error signals will be in the form of separate synchro rotor output in accordance with Specification MIL-I-23366. The output voltage shall be zero for zero heading error and zero course error. If the actual heading is to the left of the selected heading and course the Voltage "H" prime to "C" shall be in phase with the power excitation ground to phase. If the actual heading is to the right of the selected heading and course the Voltage "C" to "H" prime shall be in phase with the power excitation ground to phase. The loading of these input signals shall be 10,000 ohms or greater. The error signal gradient shall be 200 millivolts per degree.

3.4.3.3 Radio Signal Inputs - The localizer deviation and flag and omni-range signals will be from a Receiver R-541/ARN-14D conforming to Specification MIL-R-8019. The glide slope deviation and flag signals will be from a Radio Receiving Set AN/ARN-18 conforming to Specification MIL-R-25497. The Tacan deviation and distance signal shall be from a type Radio Set AN/ARN-21 conforming to Specification MIL-R-7060. The loading of these radio signals shall be 1000 ohms by the flight director computer.

3.4.3.4 Data Link Signals - The data link signals to the computer shall be obtained from a Digital Data Communication Control Set AN/ASW-13 (or equivalent) which is part of the Data Control System AN/USC-2.

3.4.4 Output Signals -

3.4.4.1 Vertical and Horizontal Pointer Output Signals - The vertical and horizontal pointer output signals shall each be that required to drive two meter movements. The signal shall be direct current with its instantaneous polarity and magnitude as required for the particular function being performed. The meter movement loads to be applied to the output signal will be driven out of view when supplied with 11 ±2.5 milliamperes. The resistance of each meter movement will be 1000 ohms ±3 percent. A deflection of 7/8 of an inch from the center position will require 2.2 milliamperes ±5 percent and will be the normal full scale command.

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3.4.4.2 Flag Signals - Each flag signal shall be that required to operate the meter movement flags having the following characteristics. The flag will be driven completely out of view with the application of 255 $^{+245}_{-0}$ microamperes during all modes of operation except ILS mode. During ILS mode the glide slope and localizer will receive the signal directly from a Radio Receiving Set AN/ARN-14 conforming to Specification MIL-R-25495 and a Radio Receiving Set AN/ARN-18 conforming to Specification MIL-R-25497. The resistance across each meter movement will be 1000 ohms ±3 percent.

3.4.4.3 Glide Slope Displacement Pointer - The glide slope displacement output signal shall drive two meter movements requiring 500 $^{+500}_{-0}$ microamperes each to be driven completely out of view. The sensitivity will be 75 microamperes per dot with a total of two dots. The load of each meter movement will be 1000 ohms ±3 percent. In the ILS mode the glide slope displacement pointer will receive the signal directly from a Radio Receiving Set AN/ARN-18.

3.5 Electrical -

3.5.1 Primary Input Power Requirements - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance from the following power sources with characteristics as defined in MIL-STD-704 having limits as specified therein.

(1) AC Power (Single Phase) 115V, Category "B", 6 VA

(2) DC Power 28V Category "B", .36 Amp

3.5.2 Wiring - The schematic wiring is shown in Figure 1. Internal wiring shall be color coded, neat and accomplished in such a manner that individual wires may be easily traced. Wires shall be secured mechanically to the terminals prior to soldering, where practicable. The wiring shall be insulated from the instrument case.

3.5.3 Electrical Receptacle - The electrical receptacles used shall conform to Specification MIL-C-26482 and to MS3116. The type receptacle used shall be as shown in MS17870. The receptacle code number markings as specified in Specification MIL-E-5400 shall be legibly and permanently applied to the case or receptacle of the instrument.

3.6 Performance - The flight director computer shall perform satisfactorily when subjected to the tests specified in Section 4 of this specification.

3.6.1 Reliability - The flight director computer shall be designed to meet the requirements of Specification MIL-R- 22973.

3.6.2 Reliability in Mean Time Between Failures - The instrument shall have a minimum of 1000 hours of mean (operating) time between failures when tested and accepted as outlined under the requirements of 4.4.3.

3.7 Identification -

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3.7.1 Nameplate - A nameplate conforming to Specification MIL-N-18307 shall be securely attached to the exterior of the case and shall be durably marked to withstand useage encountered in service. The nameplate shall be legibly marked in accordance with MIL-STD-130.

3.7.2 Manufacturer's Part Number - The manufacturer's part number on the name plate shall be identical with the manufacturer's engineering production drawing number including applicable dash numbers if the drawing is tabulated and covers more than one part.

3.8 Interchangeability - All like parts shall be directly and completely interchangeable with each other in accordance with Specification MIL-E-5400. Changes in manufacturer's part number shall be governed by the drawing number requirements of Specification MIL-D-70327.

3.9 System Operation - The computer specified herein shall operate satisfactorily as part of the integrated flight instrument system meeting the requirements of specification MIL-S-23535.

3.10 Installation -

3.10.1 Installation Instructions - The contractor shall furnish with each instrument one copy of the installation instructions with illustrations and diagrams if necessary. Prior to printing, two copies shall be furnished to the procuring activity for approval. Whenever possible the instructions shall be arranged to require only one sheet of paper either 8 1/2 by 11 or 11 by 17 inches. The instructions shall be inserted in an envelope furnished by the contractor, marked "This Envelope Contains Installation Instructions" and packaged with each instrument.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's Responsibility - The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract. The Government reserves the right to perform any of the inspections set forth in the specification when such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of Tests - The inspection and testing of the flight director computer shall be classified as follows:

(1) Preproduction Tests: Preproduction tests are those tests performed after award of contract on sample instruments that are representative in design, performance and configuration of the item which will be produced on the manufacturer's production line to determine that the instrument meets the requirements of this specification.

(2) Acceptance Tests: Acceptance tests are those tests performed on instruments manufactured and submitted for acceptance under contract.

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4.3 Preproduction Tests - The preproduction tests shall consist of all the tests of 4.6 of this specification and the preproduction tests of Specification MIL-S-23535.

4.3.1 Preproduction Test Sample Instructions- Preproduction test samples shall consist of three computers manufactured in accordance with the specification. The instruments submitted for preproduction testing shall not have been previously tested except for the Individual Tests. The samples shall be forwarded to the testing laboratory designated by the procuring activity.

4.3.1.1 Preproduction Test Sample Identification - The test samples shall be plainly identified by securely attached, durable tags marked with the information listed below:

Sample for Preproduction Testing
Aircraft Flight Director Computer
Submitted by (Manufacturer's name, date)
for Preproduction Testing in accordance with
Specification MIL-C-23367 under contract
or order number .
MS17870- (Add proper dash number)
Manufacturer's Part Number

4.4 Acceptance Tests - The acceptance tests of the instruments shall consist of the Individual Tests, the Sampling Plan tests and the reliability tests and the tests of this specification. The contractor shall furnish all samples and shall be responsible for accomplishing the required tests except that the Sampling Plan B tests shall be conducted at a Government Laboratory designated by the procuring activity. Acceptance tests except for Sampling Plan B tests, shall be under the supervision of the Government Inspector. The contractor shall furnish test reports showing quantitative results for all tests required by this specification, signed by an authorized representative of the contractor or laboratory as applicable. Acceptance or approval of material during the course of manufacture shall in no case be construed as a guarantee of the acceptance of the finished product.

4.4.1 Individual Tests - The Individual Tests of the computer shall consist of the following tests conducted on each computer:

Inspection	ILS mode
Dielectric strength	ILS approach mode
Leakage	ILS manual
Signal adjustments	Radio track mode
Auto-Nav mode	Data link
Manual heading mode	

4.4.2 Sampling Plan Tests - The Sampling Plan Tests shall consist of Sampling Plan A and Sampling B tests. The test samples selected for sampling tests shall first have passed the individual tests. The test samples which have been subjected to Sampling Plan A Tests shall not be delivered on contract until they have been refurbished and resubmitted and passed all the Individual Tests. Test samples which have been subjected to the Sampling Plan B Tests shall not be delivered on Contract.

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4.4.2.1 Sampling Plan A Sample Selection - Sampling Plan A samples shall be selected at random in accordance with the following schedule:

Quantity of Instruments Offered for Acceptance		Quantity Selected for Testing
First	15	*
Next	50	1
Next	75	1
Next	100	1
Each additional 200 or fraction thereof		1

*NOTE: When Sampling Plan B is invoked the quantity is 0. When B is omitted the quantity is 1.

4.4.2.1.1 Sampling Plan A Tests - Each instrument selected for Sampling Plan A testing shall be subjected to the tests listed below, conducted in the order listed:

Vibration error
Voltage and frequency variation
Low temperature operation
High temperature operation

4.4.2.2 Sampling Plan B Instructions - Three instruments shall be selected at random from the first fifteen (15) produced on contract and submitted within 10 days after manufacture. These instruments shall be forwarded to a Government Laboratory designated by the procuring activity. Each sample shall be plainly identified by a securely attached, durable tage marked with the following information:

Aircraft Flight Director Computer
Submitted by (Manufacterer's name, date)
for production acceptance Sampling Plan B
testing, in accordance with Contract /
Order No.
Manufacturer's Part Number

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4.4.2.2.1 Sampling Plan B Tests - Each instrument selected for Sampling Plan B Testing shall be subjected to the tests listed below, conducted in the order listed:

Sampling Plan A Tests	Temperature shock
Radio noise interference	Vibration endurance
High temperature exposure	Acceleration
High altitude-low temperature	Life

4.4.3 Reliability Testing -

4.4.3.1 Mean Time Between Failures (MTBF) Determination - Tests shall be performed to determine the MTBF of the computer. Specification MIL-R-22973 shall be used as a guide for the determination. The procedure is outlined below.

4.4.3.2 Test Procedure Outline - A Detailed Reliability Test Procedure Outline shall be submitted to and approved by the procuring activity prior to the initiation of the tests. The test procedure must include but is not limited to:

(1) Computer duty cycle.

(2) Performance characteristics to be measured.

(3) Failure criteria.

4.4.3.3 Test Procedure - Three computers shall be utilized for the reliability determination which shall be performed at Test Level III of Specification MIL-R-22973. The instruments shall be tested until a total of 2000 hours has been completed or until there is a total of four failures. The tests shall be completed within four months of the date of the first delivery of the equipment on the contract. Should the selected three equipments fail to comply with the established reliability test requirements, the tests shall be repeated within thirty days on three new equipments selected by the government inspector. If a 1000 hour MTBF has not been demonstrated, acceptance of computers will be withheld until corrective action is taken and its effectiveness is demonstrated. Computers used for these tests may be refurbished and delivered upon satisfactory completion of the tests.

4.5 Test Conditions -

4.5.1 Atmospheric Conditions - Unless otherwise specified, all tests required by this specification shall be made under the following conditions:

Temperature	Room ambient 25°C ± 5°C.
Pressure	Normal atmospheric (approximately 29.92 inches Hg.).
Humidity	Room ambient up to 90% relative humidity.

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TABLE I

POTENTIOMETER	FUNCTION	RANGE	NOMINAL SETTING	TEMPERATURE EXTREMES
R1	VERTICAL POINTER CENTERING	±0.5MA	0 ±0.02	±0.05
R2	HEADING BANK LIMIT	30° to 60°	35° ±1-1/2°	±3°
R3	DATA LINK, BANK LIMIT	30° to 60°	60° ±2-1/2°	±5°
R4	TRACK BANK LIMIT	20° to 50°	35° ±1-1/2°	±3°
R5	ILS BANK LIMIT	20° to 40°	35° ±1-1/2°	±3°
R6	APPROACH BANK LIMIT	10° to 30°	15° ±0.75°	±1-1/2°
R7	ACL BANK LIMIT	10° to 30°	15° ±0.75°	±1-1/2°
R8	COURSE CUT LIMIT	20° to 50°	45° ±2°	±5°
R9	HEADING SENSITIVITY (3° INPUT)	*0.2 to 1MA	0.5±.03	±.06
R10	ACL HEADING SENSITIVITY (3° INPUT)	*0.2 to 1MA	0.5 ±.03	±.06
R11	TRACK SENSITIVITY (25° INPUT)	*0.2 to 1MA	0.5 ±.04	±.075
R12	HORIZONTAL POINTER CENTERING	0.5 MA	CENTER	CENTER
R13	APPROACH PITCH LIMIT	+15° to -15°	0 ±.5°	0 ±.5°
R14	ACL PITCH LIMIT	+15° to -15°	0 ±.5°	0 ±.5°
R15	APPROACH PITCH TRIM	5° to 20°	10° ±0.5°	±1.0°
R16	ACL PITCH TRIM	5° to 20°	10° ±0.5°	±1.0°
R17	PITCH RATE 3.33° AT .08 CPS	*0.2 to 1 MA	0.5 ±.03	±.06
R18	PITCH CANCEL (2.3° INPUT)	-.5 to +.5MA	0 ±.03	±.06
R19	PITCH ACL SENSITIVITY (3° INPUT) (ATTITUDE)	*0.2 to 1 MA	0.5 ±.03	±.06
R20	PITCH ACL SENSITIVITY (3° INPUT) (DATA LINK)	0.2 to 1 MA	0.5 ±.03	±.06
R35	INTERNAL COURSE CANCEL (7° INPUT)	-0.5 to +0.5	0 ±0.03	±.06

* FOR VOLTAGE AND FREQUENCY TESTS R9, R10, R11, R17, and R19 SHALL PROVIDE AN ADDITIONAL 10% TOLERANCE.

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4.5.2 Voltage Input - Unless otherwise specified, the computer shall be tested with 115 volt 400 cps (nominal) single phase power.

4.5.3 Properly Connected - Whenever it is specified that the computer shall be "properly connected", it shall be understood that the computer terminals are connected to a test fixture which provides the required inputs, performs the mode relay excitation connections, and provides the required loads on the output as specified in this specification.

4.5.3.1 Test Fixture Inputs To Computer - The input from the test fixture to the computer shall simulate the radio signals required by the computer, the pitch and roll attitude signals, the heading and course error signals.

4.5.3.1.2 Test Fixture Roll, Pitch, Heading Error, and Course Error Signals - The roll, pitch, heading error, and course error signals shall be from standard test transmitters as specified herein. The test transmitters shall be manually settable to fixed displacements and shall also provide a 0.08 cps oscillation when testing the rate circuits.

4.5.3.2 Test Fixture Loads On Computer - Two 1000 ohm loads in parallel shall be utilized on each output signal.

4.5.3.3 Test Fixture Mode Relay - The test fixture shall provide switching such that the various mode relays can be energized individually or in the combinations listed in Section 3 of this specification.

4.5.4 Standard Synchro Test Transmitter - A standard calibrated synchro with minimum and known errors, shall be an Eclipse-Pioneer Type AY201-1, or equivalent, high precision transmitting synchro. A 0 to 360 degrees dial clamped to the rotor of the synchro shall be settable, rotatable, and readable through 360 degrees of rotation to 6 minutes of arc. The following procedure shall be used to calibrate the test synchro and properly identify the synchro rotor and stator leads.

(1) Designate the rotor leads as "H" and "C", to be grounded.

(2) Designate one stator lead as "Z" and connected to "C".

(3) Apply 26V, 400 cps across "H" and "C" and measure the voltage across the two remaining stator leads and the voltage across "H" and one of the stator leads.

(4) Turn the rotor until the voltage across "H" and the one stator lead is approximately maximum (about 35 volts) and the voltage across the two stator leads is exactly minimum. Lock the dial to the rotor with the dial exactly on zero.

(5) Turn the dial through increasing numbers and measure the voltage across the two undesignated stator leads and "Z". The voltage across "XZ" will increase before decreasing and across "YZ" will decrease before increasing.

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4.6 Test Methods -

4.6.1 Examination of the Product - The computer shall be examined externally to determine conformance to the applicable drawings and to all the requirements of this specification not covered by tests.

4.6.2 Dielectric Strength Test - A potential of 200 volts 60 cps shall be applied between isolated pins and the case for a period of 5 seconds. There shall be no breakdown of insulation or any other permanent damage to the instrument as a result of this test.

4.6.3 Leakage Test - Each computer case shall be tested for case leakage with a mass spectrometer or an equivalent, at least equally sensitive, quantitive leak test. The gas or mixture of gasses in the case, adjusted to a pressure differential of one atmosphere, shall not have a leakage rate from the case, exceeding one micron cubic foot per hour (10^5 cubic centimeters per second).

4.6.4 Signal Adjustment Test - The computer shall be properly connected with all input signals initially set at zero signal or displacement. Each of the adjustments of Table I shall be checked for range and specified setting.

4.6.5 Auto-Nav Mode Test - (1) With no external 28 volt source supplied to the computer, the following functions shall be displayed:

(a) Vertical steering pointer - out of view to right.

(b) Horizontal steering pointer - out of view to bottom.

(c) Glide slope displacement bar - out of view to top.

(d) Localizer warning flag - out of view.

(e) Glide slope warning flag - out of view.

(2) The computer shall be properly connected with power applied and all input signals initially set at zero signal or displacement. The operation of the computer shall be that specified above.

4.6.6 Manual Heading Mode Test - The flight director shall be placed in this mode when relay K1 and K4 are excited with 28 volts direct current from an external source to pins "J" and "E" as shown in Figure 1. The following functions shall be displayed:

(1) The vertical steering pointer shall receive a signal of heading error and bank angle. The correct zeroing action shall cause the desired heading to be achieved. The amount of called for bank angle shall be limited. All sensitivities and limits shall be in accordance with Table I.

(2) All other pointers shall be out of view as given in the Auto-Nav mode test.

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4.6.7 ILS Mode Test - The flight director shall be put in this mode of operation when relay K2 is excited with 28 volts direct current from an external source to pin K as shown in Figure 1. The following information shall be displayed on each pointer:

(1) The vertical steering pointer shall mix localizer deviation and course error localizer deviation, and bank angle through the cross wind filter (localizer rate). When the localizer deviation is high the zeroing action shall cause the course error called for to insure an intercept angle nominally set at 45°. The bank angle called for shall be limited as in the manual heading mode. The correct zeroing action shall cause an asymptotic intercept of the localizer beam. The sensitivities and limits shall be as defined in Table I.

(2) Horizontal steering pointer- out of view to bottom.

(3) Glide slope displacement bar - directly connected to glide slope receiver.

(4) Localizer warning flag - directly connected to localizer receiver.

(5) Glide slope warning flag - directly connected to glide slope receiver.

4.6.8 ILS Approach Mode Test - The computer shall be put in this mode of operation when relay K2 and K3 are excited, with 28 volts direct current from an external source to pins K and s as shown in Figure 1. The functions displayed shall be the following:

(1) The vertical steering pointer signal shall be the mixture of the following signals in the cross-wind filter of the computer: (a) localizer deviation (angular radio beam error), (b) course error rate (produced by the "MANUAL SET" heading source), (c) localizer angular rate error and (d) the bank angle rate. This signal summed in the computer with the correcting bank angle signal of the reference gyro will produce the necessary zeroing action of the needle causing the desired command heading, to be achieved as shown in Figure 1.

(2) The horizontal steering pointer signal shall be the mixture of the following signals: (a) glide slope error and altitude reference as specified in Figure 1. All sensitivities and limits shall be as shown in Table I.

(3) All other pointers shall operate as in the ILS Mode.

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4.6.9 ILS Manual Test - When relays K1, K2, and K4 are excited with 28 volts direct current from and external source to pins J, K, and E respectively, the flight director computer shall operate as follows:

(1) The vertical pointer shall mix heading error and bank angle as in the manual heading mode. The bank limit shall be the ILS limit.

(2) All other pointers shall operate as in the ILS mode.

4.6.10 Radio Track Mode Test - The flight director computer shall be placed in this mode when relay K2 is energized with 28 volts direct current from an external source to pin K. The amount of bank angle called for shall be limited. All sensitivities and limits shall be in accordance with Table I. The following functions shall be displayed:

(1) The vertical steering pointer shall receive a computed signal composed of radio information (VOR or TACAN) plus a rate signal derived from a combination of radio information (VOR or TACAN), course error VOR or TACAN, bank angle through the cross wind filter. The signal shall be such that the pointer is deflected in the direction in which control action must be taken and is centered when the aircraft is flying on the desired track or is approaching the desired track asymptotically. When the aircraft is centered on the beam or is on the desired track, cross wind components shall be compensated for, so that the desired track may be maintained.

(2) The remaining pointers shall be out of view as specified in the Auto-Nav mode test.

4.6.11 Data Link-vectoring Submode Test - The computer shall be placed in this mode when relay K4 is energized with 28 volts direct current from an external source to pin E. The pointers shall operate as follows:

(1) The vertical steering pointer shall receive a signal of heading error and bank angle. The heading error shall be the difference between the actual aircraft heading and the heading obtained from the data link vectoring control mode. The bank angle limit called for shall be as specified in Table I.

(2) All other pointers shall be out of view as specified in the Auto-Nav mode test.

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4.6.11.1 Data Link-traffic Control Submode Test - The computer shall be placed in this mode when relay K4 is energized with 28 volts direct current from an external source to pin E. The pointers shall operate as follows:

(1) The vertical steering pointer shall receive a signal of heading error and bank angle. The heading error shall be the difference between the actual aircraft heading and the heading obtained from the data link traffic control mode. The bank angle limit called for shall be as specified in Table I.

(2) All other pointers shall be out of view as specified in the Auto-Nav mode test.

4.6.11.2 Data Link-Automatic Carrier Landing (ACL) Submode Test - The computer shall be placed in this mode when relay K4 and K7 are energized with 28 volts direct current from an external source to pins E and t. The pointers shall operate as follows:

(1) The vertical steering pointer shall receive a signal of heading error and bank angle. The heading error shall correspond to the bank angle command obtained from the data link ACL mode. The bank angle limit called for shall be as specified in Table I.

(2) The horizontal steering pointer shall receive a signal of pitch attitude error and pitch attitude. The pitch attitude error shall correspond to the pitch angle command obtained from the data link ACL mode. The pitch attitude limit called for shall be as specified in Table I.

(3) All other pointers shall be out of view as specified in the Auto-Nav mode test.

4.6.11.3 Data Link-Precision Course Direction Submode Test - The computer shall be placed in this mode when relay K4 is energized with 28 volts direct current from an external source to pin E. The pointers shall operate as follows:

(1) The vertical pointer shall receive a signal of heading error and bank angle. The heading error shall be the heading obtained from the data link precision course direction mode. The bank angle called for shall be as specified in Table I.

(2) All other pointers shall be out of view as specified in the Auto-Nav mode test.

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4.6.12 Vibration Error Test - The computer shall be properly connected with power applied. The computer with shock mount shall be mounted on a vibration table in a position to simulate service installation orientation. The computer shall be subjected to a vibration test conducted with the frequency cycling between 10 and 500 cps in a 15-minute cycle, at an applied double amplitude of 0.010 inch or an applied acceleration of ±2g, whichever is the limiting value. There shall be no deviation in the outputs during vibration for any mode of operation.

4.6.13 Voltage and Frequency Variation Test - The computer shall be properly connected. The input voltage and frequency shall be varied within the following limits and combination:

Frequency	380-420 cps
AC Voltage	103-127 volts
DC Voltage	22 volts

The computer shall be subjected to and meet the requirements of the mode tests described in 4.6.6 to 4.6.11.3 inclusive.

4.6.14 Low Temperature Operation Test - The computer shall be properly connected except that no power shall be supplied, and shall be subjected to a temperature of -54°C (-65°F) for a period of 4 hours. At the end of the 4-hour period and while still at the low temperature, power shall be applied and the computer shall be subjected to and meet the requirements of the tests listed under Voltage and Frequency Variation Test. Immediately following removal from the temperature chamber, the computer shall be subjected to and meet the requirements of the Leakage test.

4.6.15 High Temperature Operation Test - (This test may be combined with the High temperature exposure test at the discretion of the testing agency.) The computer shall be properly connected, except that no power shall be applied and shall be subjected to a temperature of 71°C (160°F) for a period of 4 hours. At the end of the 4-hour period and while still at the high temperature, power shall be applied and the computer shall be subjected to and meet the requirements of the tests listed under Voltage and Frequency Variation Test. Immediately following the removal from the temperature chamber, the computer shall be subjected to and meet the requirements of the Leakage test.

4.6.16 Radio Noise Interference Test - The electrical system of the computer shall be tested for conducted and radiated radio noise interference in accordance with Specification MIL-I-6181.

4.6.17 High Temperature Exposure Test - The computer shall be properly connected except that no power shall be applied. The computer shall then be tested in accordance with Procedure II, high temperature tests, of Specification MIL-E-5272 for a period of 24 hours. At the end of this period, and while still at the high temperature, the computer shall be subjected to and meet the requirements of the mode tests described in 4.6.6 to 4.6.11.3 inclusive. After the computer has returned to room temperature, it shall be subjected to and meet the requirements of all individual tests.

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4.6.18 Low Temperature-High Altitude Test - The computer shall be properly connected except that no power shall be applied. The computer shall then be tested in accordance with Procedure II, low temperature tests, of Specification MIL-E-5272. At the end of the specified period, and while still at the low temperature, the pressure in chamber shall be reduced to the equivalent of 80,000 ± 500 feet and power applied. The computer shall be subjected to and meet the requirements of the mode tests described in 4.6.6 to 4.6.11.3 inclusive. The pressure in the test chamber shall be increased to atmospheric and after the computer has returned to room temperature, it shall meet the requirements of all the individual tests.

4.6.19 Temperature Shock Test - The computer shall be tested in accordance with Procedure I, temperature shock tests, of Specification MIL-E-5272. After completion of the test, the computer shall be subjected to and meet the requirements of the individual tests.

4.6.20 Vibration Endurance Test - The computer shall be tested in accordance with Procedure XII of Specification MIL-E-5272. After these vibration tests, the computer shall be properly connected and shall be subjected to and meet the requirements of the individual tests. In addition, no looseness nor damage to any part of the computer shall result from the vibrations to which the components were subjected.

4.6.21 Acceleration Test - The computer, not operating, shall be tested in accordance with Procedure III acceleration test of Specification MIL-E-5272 except that the computer shall be subjected to an acceleration of 20g. At the end of this acceleration the computer shall be subjected to and meet the requirements of the individual tests.

4.6.22 Life Test - The computer shall be properly connected with power applied. The test shall last for 1000 hours, and each relay in the computer shall be cycled 20 times per hour. There shall be no failure during the test. The test may be interrupted at 250-hour intervals to determine compliance with part or all of the individual tests. There shall be no failure during the test. After the 1000 hours operation, the computer shall be subjected to and meet the requirements of the individual tests. The testing agency may run this life test on three hour cycle, 2-1/2 hours on and one-half hour off, in which case hours of life shall be the sum of the periods.

4.6.23 Humidity Test - The computer shall be tested in accordance with Procedure I of Specification MIL-E-5272. At the end of this test the computer shall be subjected to and meet the requirements of the mode tests described in 4.6.6 to 4.6.11.3 inclusive.

4.6.24 Fungus Resistance Test - The computer shall be tested in accordance with Procedure I of Specification MIL-E-5272, except that the test period shall be 14 days. At the option of the testing agency, tests of component parts of the item may be accepted in lieu of, or in addition to, tests of the assembled instrument.

4.6.25 Salt Spray Test - The computer shall be properly connected except that no power shall be applied. The computer shall then be tested in accordance with Procedure I, salt spray test, of Specification MIL-E-5272. At the end of the 50-hour period, the computer shall be subjected to and meet the requirements of the individual tests.

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4.7 Rejection and Retest - Instruments which have been rejected may be reconditioned or have parts replaced to correct the defects and re-submitted for acceptance. Before resubmission, full particulars concerning previous rejection and action taken to correct the defects shall be furnished the government inspector.

5. PREPARATION FOR DELIVERY

5.1 Packaging -

5.1.1 Level A - The computer shall be packaged in accordance with Specification MIL-P-7936, Level A or B as specified in the contract or order (see 6.2). The method of preservation shall be in accordance with Specification MIL-P-116, Method IA5 omitting preservation compound and using reusable metal containers.

5.1.2 Level C - When this level is required, packaging shall be in accordance with standard commercial practice.

5.2 Packing -

5.2.1 Levels A and B - The instrument shall be packed in accordance with Specification MIL-P-7936. The level or levels required shall be as specified in the contract or order (see 6.2).

5.2.2 Level C - The instrument shall be packed to insure that the shipment arrives in a satisfactory condition at destination. The shipment shall conform to the applicable carriers rules and regulations in effect at the time of shipment.

5.3 Marking - The interior and exterior containers shall be marked as specified in Specification MIL-P-7936.

5.3.1 Precautionary Marking - The following precautionary markings shall appear on two opposite sides of each interior package and shipping container whenever practicable, depending on the size of the carton:

FRAGILE
DELICATE INSTRUMENTS
HANDLE WITH CARE

6. NOTES

6.1 Intended Use - The flight director computer covered by this specification is intended for use in high performance aircraft as part of an integrated flight instrumentation system. It will provide the computed flight director signals to the attitude director indicators.

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6.2 Ordering Data - Procurement documents should specify the following:

(1) Title, number and date of this Specification.

(2) The quantity, MS part number (and type number, if any) of the instrument desired.

(3) Levels of packaging (see 5.1) and packing (see 5.2) desired.

6.3 Drawing Requirements - The attention of contractors is invited to the fact that Government contracts for equipment, whether procured directly by the Government or through another contractor, require that all drawings submitted must be in accordance with the requirements of Specification MIL-D-70327.

6.4 Definitions -

6.4.1 Amplitude - Whenever the word "amplitude" is specified, it shall mean the extent of motion as measured from one extreme to the opposite extreme.

6.4.2 Hermetic Seal - A hermetic seal shall be a perfectly closed and air tight seal made between metallic and metallic or between metallic and vitric materials. A hermetic seal is not intended to include seals accomplished by gaskets.

6.5 Preproduction Test Provisions - The manufacture of instruments on contract shall not commence until the samples submitted are pronounced satisfactory by the procuring activity. When a contractor is in continuous production of the instrument from contract to contract, the submission of further preproduction samples on the subsequent contracts may be waived at the discretion of the procuring activity. Approval of preproduction samples or the waiving of preproduction tests does not preclude the requirement of acceptance testing.

6.5.1 It shall be understood that the instrument supplied under contract or order shall be identical to the corresponding preproduction sample in design, construction, quality, material, workmanship and method of manufacture. Deviation from the standards of the preproduction sample shall be made only by the procuring activity. Evidence of unauthorized change shall constitute cause for rejection.

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